

1 ☆

QUESTION: 1 GROUP: Aptitude SECTION: Puzzles

Mark(s) 1

What will be the next term of the series?

-1100, -810, -576, -392, —

☐ -252☒ -256 ✓☐ -225☐ -300

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QUESTION 2

GROUP

Aptitude

SECTION

Problem Solving

The ages of Mohit and Amit differ by 30 years. 5 years later, Mohit is 6 times as old as Amit. What will be the present age of Amit?

☐ 6 years☐ 7 years☐ 5 years

1 year

 Unanswer

Mark(s): 1

The ages of Mohit and Amit differ by 30 years. 5 years later, Mohit is 6 times as old as Amit. What will be the present age of Amit?

☒ 1 year

☐ 7 years

☐ 5 years

☐ 6 years

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Mark(s): 1

The percentage of sodium in solution A is twice the percentage of sodium in solution B. 60% of solution B is nitrate. If solutions A and B are mixed in the ratio of 4 : 3, the percentage of sodium in the final mixture? Assume that both solutions have only sodium and nitrate.

☒ 62.8%☐ 37.5%☐ 40%☐ 48.2%[Previous Question](#)[Next Question](#)

Mark(s) 1

If + means $\times$, - means + and $\times$ means - then solve the below expression:

$$3 + 50 - 170 \times 120$$

☒ 200

☐ -20357

☐ 100

☐ 220

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Direction: The question is followed by two statements, I and II. Choose the statement/statements required to answer the question by selecting a suitable option.

Mark (A) if the question can be answered by one of the statements alone but not by the other.

Mark (B) if the question can be answered using either statement alone.

Mark (C) if the question can be answered using both statements together, but can not be answered using either statement alone.

Mark (D) if the question cannot be answered even by using both statements together.

Question: There are 5 even consecutive numbers, if they are arranged in ascending order, find the 2nd number.

I) The second largest number is 150% of the smallest number.

II) The difference between the 3rd largest number and 4th smallest number is 18.

☐ (C)

☐ (D)

☐ (A)

☒ (B)

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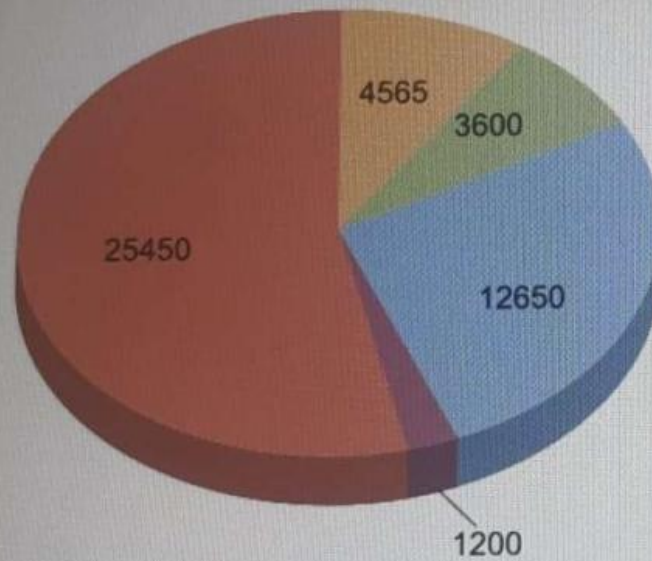
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Mark(s): 1

Direction: Study the information given below and answer the question that follows.

The Pie chart shows the funds to be collected (in crores of rupees) from different sources by the Airport Authority of India (AAI) for their projects of development.

- Market Borrowing
- External Assistance
- Annuity
- Government Aid
- Central Bank



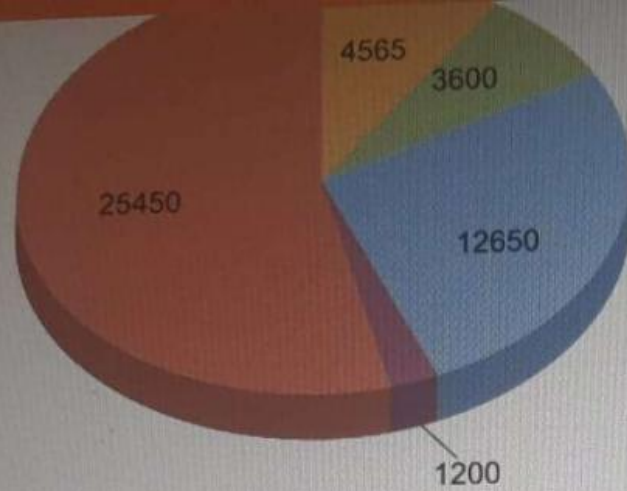
Question: If the Annuity is to be collected through an outsourced agency by allowing a maximum 20% commission, how much amount should be permitted to be collected at the project is supported with Rs. 12650 crores?

☐ Rs 2530 crores

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● Government Aid
● Central Bank



Question: If the Annuity is to be collected through an outsourced agency by allowing a maximum 20% commission, how much amount should be permitted to be collected that the project is supported with Rs. 12650 crores?

- ☐ Rs 2530 crores
- ☐ Rs 12650 crores
- ☐ Rs 3036 crores
- ☒ Rs 15180 crores

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QUESTION: 11 GROUP: Aptitude SECTION: Puzzles

Mark(s) 1

Jaya starts at point A, walks straight to point B which is 5 ft away. She turns right at 90° and walks to C which is 5 ft away, turns 90° left and goes 4 ft to D, turns 90° left and goes to E, which is 2 ft away and once again turns 90° left and goes to G, 3 ft away. What is the distance between A and G?

☐ 15 ft

☐ 13 ft

☐ 14 ft

☐ 11 ft

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QUESTION: 13 GROUP: Aptitude SECTION: Problem Solving

Mark(s): 1

If $N = (1 \cdot 2 \cdot 3 \cdot \dots \cdot 88) / 8^x$, what is the least value of x for which N is not an integer?

☐ 28

☐ 26

☒ 29

☐ 27

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QUESTION: 15 GROUP: Aptitude SECTION: Puzzles

Mark(s): 1

In three of the given four options, the 2nd number is related to the 1st number in the same logical way. Which is the odd one out?

☐ 423 - 29

☐ 321 - 12

☐ 210 - 5

☐ 522 - 33

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QUESTION 23 GROUP: Programming SECTION: C Programming

Mark(s): 1

18

What would be the size of "storearray" in reference to the code snippet given below?

19

```
short storearray[4][3] = { {3}, {4, 6}, {2, 6, 9}};
```

20

21

☐ 7

22

☐ 18

23 ☆

☒ 24 ✓

24

☐ 12

25

26

27

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Mark(s): 1

Which one of the statement given in options is true about the program given below?

```
#include <iostream>
using namespace std;

class A
{
    int b;
public:
    A()
    {
        cout << "CVB" << endl;
    }
};

int main()
{
    class A
    {
        int a = 3;
public:
```

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QUESTION 25 GROUP Programming SECTION: CPP Programming

Mark(s) 1

What will be the output of the program given below?

```
#include <iostream>
#include <fstream>
using namespace std;

struct Stud
{
    int a[10];
};

int main()
{
    Stud s = { 3, 5, 8, 6 };
    ofstream O;
    int *p = new int[5];
    O.open ("num.txt", ios::out);
    O.write((char *) s.a, 5 * sizeof(int));
    O.close();
    ifstream t;
    t.open("num.txt");
    t.read((char *)p, 5 * sizeof(int));
    t.close();
    for (int i = 0; i < 4; i++)
```

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```

17  ofstream O;
    int *p = new int(5);
    O.open ("num.txt", ios::out);
    O.write((char *) p, 5 * sizeof(int));
18  O.close();
19  ifstream t;
    t.open("num.txt");
20  t.read((char *)p, 5 * sizeof(int));
    t.close();
21  for (int i = 0; i < 4; i++)
        cout << p[i] << " ";
22  return 0;
23  }

```

☐ 3 5 8 6

☐ 3 5 8 0

☐ 0 0 0 0

☐ 3 garbage values

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QUESTION 26 GROUP: Programming SECTION: C Programming
Mark(s): 1

Consider the following

I) _LINE_

II) _TIME_

III) _FILE_

IV) _DATE_

V) _STDC_

Which of the following is true?

☐ Only (I) and (II) is predefined macro in C

☒ All are predefined macro in C ✓

☐ Only (IV) and (V) is predefined macro in C

☐ Only (II) and (III) is predefined macro in C

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ASUS VivoBook

17 QUESTION: 27 GROUP: Programming SECTION: C++ Programming
18 Mark(s) 1

19 What happens when a member function throws an exception?

- 20 ☐ The program terminates.
- 21 ☐ Control is transferred to the catch block following the try block.
- 22 ☐ Program returns a run time error.
- 23 ☐ Control returns to the calling function.
- 24
- 25
- 26

27 ☆

28

29

30

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QUESTION 28 GROUP: Programming SECTION: Python Programming
Mark(s) 1

Predict the output of the following Python program:

```
import numpy as np
x = np.arange(12).reshape((2, 6))
r1 = np.ptp(x, 1)
r2 = np.amax(x, 1) - np.amin(x, 1)
assert np.allclose(r1, r2)
print(r1)
```

☐ [6 6]

☒ [5 5]

☐ [5 6]

☐ [6 5]

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QUESTION 29 GROUP: Programming SECTION: Python Programming
Mark(s) 1

What will be the output of the following Python Program?

```
x = ["apple", "banana"]
```

```
print("pineapple" not in x)
```

☒ True

☐ Banana

☐ Apple

☐ False

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What will be the output of the program given below?

```
#include <stdio.h>
int main()
{
    int arr[2][3][10] = {2, 4, 6, 8, 5, 7};
    int *p = (int *) &arr;
    printf("%d %d %d", *p++, *(*(arr + 2) + 1) + 1), ++*p++;
    return 0;
}
```

☐ Garbage values

☒ 4 Garbage value 3

☐ 2 6 4

☐ 2 0 6

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QUESTION 31 GROUP: Technical SECTION: Analog Circuits

Mark(s) 1

For a transistor in CB configuration, the collector current is 4.5mA and base current is 45uA. Calculate the value of current amplification factor.

0.97

0.99

0.95

0.98

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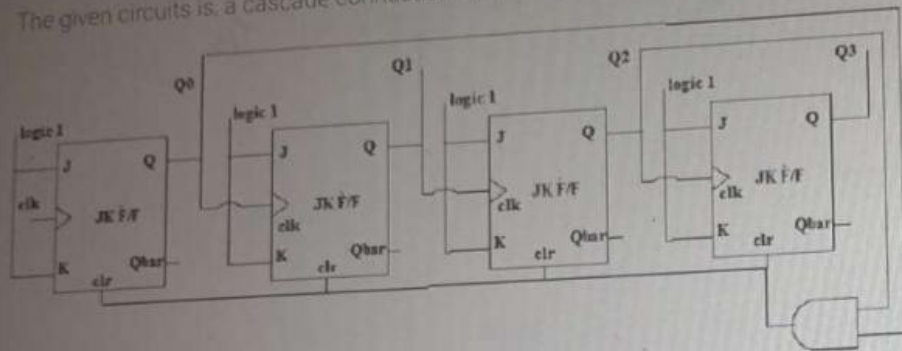
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QUESTION 32 GROUP: Technical SECTION Digital Electronics

Mark(s): 1

32 ☆

The given circuits is a cascade connection of 4 J-K flip flop which works as:



- ☐ mod 10 asynchronous UP counter
- ☐ mod 10 asynchronous DOWN counter
- ☐ mod 5 asynchronous UP counter
- ☐ mod 5 asynchronous DOWN counter

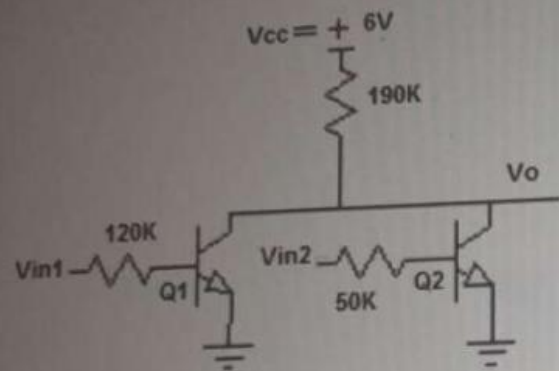
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QUESTION 33 GROUP Technical SECTION RTL
Mark(s) 1

A circuit diagram of RTL is shown in figure, with $V_{CC} = 6V$



Determine the output of the following circuit when V_{in1} & V_{in2} both are at logic 0-

☐ $V_o = 3V$

☐ $V_o = 1V$

☐ $V_o = 0V$

☐ $V_o = 6V$

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Determine the value of conductance band density for a semiconductor material?

Given that Band gap energy and KT ratio is 1.11,

n_i is $2.7 \times 10^{11} \text{ cm}^{-3}$ when temperature is 1000K

Valence band density is $1.8 \times 10^{18} \text{ cm}^{-3}$.

☐ $N_c = 0.594 \times 10^4 \text{ cm}^{-3}$

☐ $N_c = 3.594 \times 10^4 \text{ cm}^{-3}$

☐ $N_c = 2.554 \times 10^4 \text{ cm}^{-3}$

☐ $N_c = 1.524 \times 10^4 \text{ cm}^{-3}$

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Which of the following is not the type of sequential circuit in VLSI design?

- ☐ Monostable circuits like timers
- ☐ Tristable circuits like tristate switches
- ☐ Astable circuits like ring oscillator
- ☐ Bistable circuits like flip flop and registers

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Mark(s): 1

Which two registers share same memory location when 8051 is powered up?

☐ R1 and TH

☐ R0 and SP

☐ R3 and DPL

☐ R2 and TL

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Which of the below given electronic device produces a magnetic field when a constant electric current run through it?

☐ Conductor

☐ Inductor

☐ Transistor

☐ Diode

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Which of the following types match the characteristics of FPGA?

☐ SRAM

☐ PLD

☐ EPROM

☐ SLD

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Determine the high level output voltage for an RTL gate for a fanout of 5, with $V_{cc} = 3.6\text{ V}$, $R_B = 450\ \Omega$, $R_C = 640\ \Omega$.

☐ 1.23 V

☐ 1.98 V

☐ 1.56 V

☐ 1.058 V

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Which of the Computer Aided Design softwares employed in VLSI design automation?

- ☐ CADENCE only
- ☐ CADENCE and SYNOPSYS only
- ☐ CADENCE, SYNOPSYS, MENTOR and MATHWORKS
- ☐ CADENCE, SYNOPSYS and MENTOR only

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Mark(s): 1

In 4 Bit Ripple counter, JK flip flops are used. If the propagation delay of each JK Flip Flop is 8 ns, what will be the worst case propagation delay for the 4 Bit Ripple Counter?

☐ 32 ns

☐ 40 ns

☐ 8 ns

☐ 16 ns

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Mark(s): 1

For an FET if Drain Source voltage is 4V and I_D is 2mA, what is the maximum power dissipated?

☐ 2 mW

☐ 4 kW

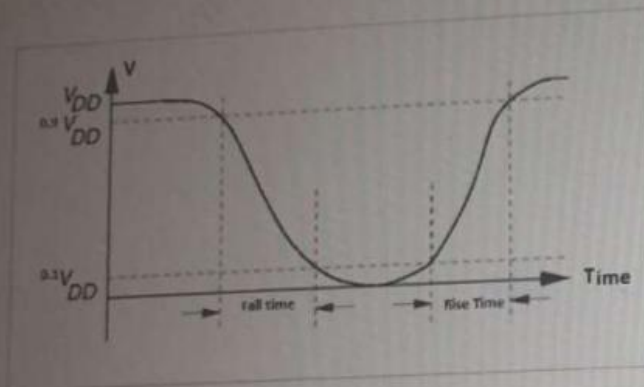
☐ 2 kW

☐ 8 mW

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Rise and Fall Time characteristics as shown below are obtained while carrying out testing.



This is generally obtained during which type of testing

- ☐ AC Parametric Test
- ☐ DC Parametric Test
- ☐ SCAN
- ☐ BIST

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Mark(s): 1

How many memory cycles are required to execute the following two instructions?

LDA 1200H

LXI D, 01F2H

☐ 8

☐ 4

☐ 7

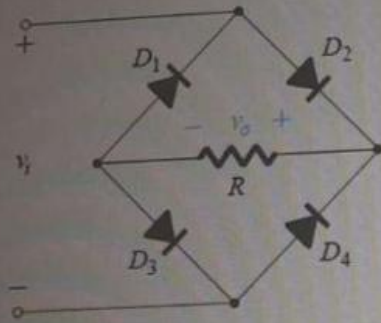
☐ 6

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QUESTION 48 GROUP: Technical SECTION: Analog Circuits
Mark(s) 1

If V_i is having 100Hz frequency, what will be frequency of V_o ?



- ☐ 100Hz
- ☐ 25Hz
- ☐ 50Hz
- ☐ 150Hz

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What is the difference between Latch & Flip Flop?

Latches are Level Triggered and Flip Flop are edge Triggered.

Latches are used in synchronous circuit and Flip Flop is used in Asynchronous circuit.

Latches are used in Analog circuit and Flip Flop is used in Digital circuit.

Latches are used to store bits and Flip Flop is used to store data.

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What is the insertion ratio of a 2 port network?

Ratio of power transferred from source to load before, and after insertion of 2 port network between source and load.

Ratio of power loss due to insertion, and power transferred from source to load before insertion of 2 port network between source and load.

Ratio of power transferred from source to load after insertion of 2 port network between source and load, and power loss due to insertion.

Ratio of power transferred from source to load after, and before insertion of 2 port network between source and load.

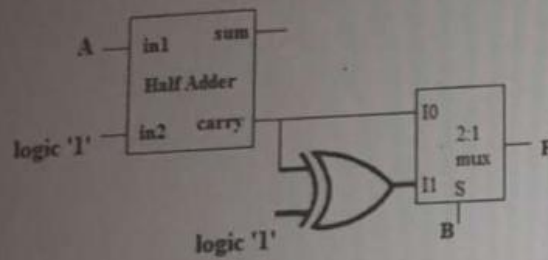
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QUESTION: 52 GROUP: Technical SECTION: Analog Circuits

Mark(s): 1

The circuit shown below uses one XOR gate, one Half Adder & one 2:1 multiplexer circuit. The inputs are A & B while the output if F realizes an



☐ $F = \bar{A} + B$

☐ $F = \overline{A \oplus B}$

☐ $F = B$

☐ $F = A \oplus B$

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QUESTION 53 GROUP: Technical SECTION: RTL

Mark(s) 1

For a RTL timing operation, relation between cyclic time and longest critical path?

- ☐ cyclic time less then longest critical path
- ☐ cyclic time greater then longest critical path
- ☐ There is no relation between cyclic time and longest critical path.
- ☐ cyclic time equal to longest critical path

Previous Question

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QUESTION 54 GROUP: Technical SECTION: Verilog

Mark(s) 1

VMM Library has which of the following sub library?

- ☐ Standard Library and Register Abstraction Layer only.
- ☐ Register Abstraction Layer and Hardware Abstraction Layer.
- ☐ Hardware Abstraction Layer only.
- ☐ Standard Library, Register Abstraction Layer, Hardware Abstraction Layer and Score boarding

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QUESTION 55 GROUP Technical SECTION Verilog

Marks: 1

Find the result of the given construct if passed through synthesis tool.

`a = #40 b;`

- ☐ a=b
- ☐ Relational operation is performed
- ☐ 40 is stored in 'a'
- ☐ Error occurred

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ASUS VivoBook